LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Physics Formula & Rules Sheet (Class 10)

Light: mirrors & lenses

What	Formula / rule	Watch out
Mirror formula	$1/v + 1/u = 1/f$	u is always negative
Lens formula	$1/v – 1/u = 1/f$	Minus sign, unlike the mirror
Magnification (mirror)	$m = h'/h = –v/u$	–ve m = real & inverted
Magnification (lens)	$m = h'/h = v/u$	+ve m = virtual & erect
Focal length of a mirror	$f = R/2$	Concave f –ve, convex f +ve
Power of a lens	$P = 1/f$ (f in metres), unit dioptre (D)	Convex +, concave –. Convert cm → m!
Lenses in contact	$P = P_1 + P_2 + \ldots$	Add with signs
Refractive index	$n = c/v \cdot n_{21} = v_1/v_2 \cdot n = \sin i / \sin r$	No unit; $c = 3 \times 10^8$ m/s

Sign convention: measure from the pole or optical centre. The object is on the left. Along the incident light is +, against it is –. Above the axis is +, below is –.

- **Ray rules:** parallel ray → through F (or appears to come from F). Ray through F → parallel. Ray through C (mirror) → returns on itself. Ray through O (lens) → straight on.
- **Same-size real image:** object at C (concave mirror) or at $2F_1$ (convex lens).
- **Virtual & enlarged:** object between P and F (concave mirror) or between F_1 and O (convex lens).
- **Always virtual, erect, diminished:** convex mirror and concave lens.

Human eye

Defect	Image forms	Lens
Myopia (can't see far)	In front of the retina	Concave (–ve power)
Hypermetropia (can't see near)	Behind the retina	Convex (+ve power)
Presbyopia (old age)	Near point moves away	Convex / bifocal

Near point 25 cm · far point at infinity · red bends least, violet most · blue sky and red sunsets are due to **scattering** · twinkling stars and the 2-minute early sunrise are due to **atmospheric refraction**.

Electricity

Quantity	Formula	SI unit
Current	$I = Q/t$	ampere (A)
Potential difference	$V = W/Q$	volt (V) = J/C
Ohm's law	$V = IR$	ohm (Ω)
Resistance of a wire	$R = \rho L/A$	ρ in Ω m
Series	$R_s = R_1 + R_2 + R_3$	Same I everywhere
Parallel	$1/R_p = 1/R_1 + 1/R_2 + 1/R_3$	Same V across each
Heat (Joule's law)	$H = I^2Rt$	joule (J)
Power	$P = VI = I^2R = V^2/R$	watt (W)
Energy	$E = P \times t$	1 kWh = 3.6×10^6 J = 1 unit

- Electron charge 1.6×10^{-19} C, so 1 C ≈ 6.25×10^{18} electrons.
- Ammeter **in series**, voltmeter **in parallel**.
- Double the length → R doubles. Double the diameter → R becomes ¼. Stretch to double the length → R becomes 4 times.
- n equal resistors R: series gives nR; parallel gives R/n.
- A bulb on half its rated voltage uses ¼ of its rated power ($P \propto V^2$ for fixed R).

Magnetic effects

- **Right-hand thumb rule:** thumb along the current → curled fingers show the field.
- **Fleming's left-hand rule: F**orefinger = **F**ield, **C**entre (middle) finger = **C**urrent, thu**M**b = **M**otion (force).
- Force is greatest when the current is ⊥ to the field, and zero when parallel to it.
- Domestic wiring: live **red** · neutral **black** · earth **green** · 220 V · 15 A circuit for heavy appliances, 5 A for lights and fans · fuse in the live wire.